Band-1 Map Folding: 2×n Strip Increment (OEIS A001415)

Finite box certificates + A001415 ledger errata (NOT asymptotics solved; NOT A001418(8))

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Document class. Scientific computational / literature ledger increment (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parent. `2026-09-11_Band1_Computational_Experiment_map_folding` (lab `/workspace/experiments/band1/m`
Lab. `/workspace/experiments/band1/map_folding_2xn/` (mission-control mirror: `.../map_folding_2xn_incr`
Cutoff. 2026-09-14.

Honesty stamp (non-negotiable).
This memo is a **finite computational / literature ledger increment** only. It **does not** claim map-folding asymptotics solved, a closed form, or a P-time algorithm. It **does not** claim A001418(8). It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims.

Abstract

We ship a Band-1 increment on rectangular map / stamp folding: (i) re-certify labeled $1 \times n$ stamp foldings OEIS A000136 for $n = 1..9$ by the independent same-parity crease non-crossing enumerator; (ii) box-certify $2 \times n$ strip foldings OEIS A001415 for $n = 0..10$ by re-executing the published Lunnon multidimensional algorithm (`mapFolding`); (iii) cross-check A000136 against `mapFolding` on the overlap and A001418 for $n = 1..4$; (iv) **correct an offset bug** in the 2026-09-11 A001415 literature table. Exact 8×8 map count A001418(8) and asymptotics remain **OPEN**.

Keywords. Map folding; stamp folding; OEIS A001415; A000136; A001418; Lunnon; Band-1; non-claim.

1. Statement lock

Definition (labeled $1 \times n$ stamp folding, A000136). Number of permutations of $\{1, \ldots, n\}$ realizable as top-to-bottom stacks of a foldable strip of n labeled stamps (Koehler / Lunnon).

Definition ($2 \times n$ strip folding, A001415). Number of ways of folding a $2 \times n$ strip of stamps along the crease grid (Lunnon $G(n, 2)$; OEIS offset 0).

Definition ($n \times n$ map folding, A001418). Number of ways of folding an $n \times n$ sheet; literature known only for $n \leq 7$.

Finite lemma (this increment only). Box re-certificates for A000136 ($n = 1..9$) and A001415 ($n = 0..10$); ledger errata for A001415. **Not** asymptotics solved; **not** A001418(8).

2. What changed vs 2026-09-11

Axis	Prior (2026-09-11)	This increment
Lab path	`/workspace/experiments/band1/map_folding/`	`/workspace/experiments/band1/map_foldi`
A000136	independent cert $n = 1..9$	**re-certified** $n = 1..9$ + mapFolding cross to $n = 12$
A001415	lit-cited only (buggy offset table)	**box-certified** $n = 0..10$ + **ledger corrected**
A001418	lit $n \leq 7$; open $n = 8$	partial box cross $n = 1..4$; $n = 8$ **still OPEN**
Claim level	finite / lit	**same; asymptotics OPEN**

3. Methods

3.1 Independent A000136 enumerator

Reuse of the parent Band-1 filter: a permutation is foldable iff no two same-parity creases $(i, i+1)$ and $(j, j+1)$ cross in stack order (Koehler 1968 / Lunnon 1968 / Wikipedia Map folding). Exhaustive over S_n for $n \leq 9$ ($9! = 362{,}880$).

3.2 A001415 via Lunnon / `mapFolding`

True geometric $2 \times n$ map folding is harder than the $1 \times n$ noncrossing filter. This increment **does not invent** a new geometric model. Instead it **re-executes** the classical Lunnon multidimensional map-folding algorithm as packaged in Hunter Hogan's `mapFolding` (PyPI; ports of Fred Lunnon C / Sean A. Irvine Java), calling `countFolds((2,n))` for $n = 0..10$, and matches OEIS A001415.

Box-certified vs lit-cited. Counts below labeled "box" are computed on-box and matched to OEIS. Larger OEIS terms ($n > 10$) and A001418 for $n = 5, 6, 7$ remain **lit-cited only** in this memo (partial A001418 $n = 1..4$ also box-crossed).

4. Unit tests

Test	Result
1324 forbidden (1×n)	PASS
all S_3 foldable	PASS
identity $n = 1..6$ foldable	PASS
A001418(2) OEIS example list length 8	PASS
ledger errata: corrected $a(1) = 2$ vs buggy 1	PASS

All unit tests PASS.

5. Re-certificates: A000136 ($1 \times n$)

Independent enumerator, $n = 1..9$:

n	Box A000136	Lit	Match	n·A000682	Wall (s)
1	1	1	Y	Y	< 0.01
2	2	2	Y	Y	< 0.01
3	6	6	Y	Y	< 0.01
4	16	16	Y	Y	< 0.01
5	50	50	Y	Y	< 0.01
6	144	144	Y	Y	0.00
7	462	462	Y	Y	0.01

n	Box A000136	Lit	Match	n·A000682	Wall (s)
8	1392	1392	Y	Y	0.11
9	4536	4536	Y	Y	1.04

Cross-check: `mapFolding.countFolds((1,n))` matches the independent counts for all $n = 1..9$, and extends with lit match through $n = 12$ (14060, 46310, 146376).

6. Headline: A001415 $(2 \times n)$ box certificates

n	Box A001415	Lit (OEIS)	Match	$4n \mid a(n)$ $(n \geq 2)$	Wall (s)
0	1	1	Y	—	< 0.01
1	2	2	Y	—	< 0.01
2	8	8	Y	Y	< 0.01
3	60	60	Y	Y	< 0.01
4	320	320	Y	Y	< 0.01
5	1980	1980	Y	Y	0.00
6	10512	10512	Y	Y	0.00
7	60788	60788	Y	Y	0.02
8	320896	320896	Y	Y	0.09
9	1787904	1787904	Y	Y	0.53
10	**9381840**	**9381840**	**Y**	**Y**	**2.43**

Headline. All eleven A001415 values $n = 0..10$ **PASS** (box = OEIS). Lunnon divisibility $4n \mid a(n)$ for $n \geq 2$ **PASS**.

7. Literature ledger refresh + errata

7.1 A001415 offset errata (important)

The 2026-09-11 Band-1 `literature_ledger.json` stored A001415 as $1, 1, 2, 8, 60, 320, 1980, 10512, 60788$ for $n = 0..8$. OEIS A001415 (offset 0) is $1, 2, 8, 60, 320, 1980, 10512, 60788, 320896$. The prior table inserted an extra 1 at $n = 1$ and left-shifted subsequent terms. **Corrected in this increment.**

7.2 A001418 $(n \times n)$ status

n	Lit A001418	This run
1	1	box cross PASS
2	8	box cross PASS
3	1368	box cross PASS

n	Lit A001418	This run
4	300608	box cross PASS
5..7	lit-known	lit-cited only here
8	**?**	**OPEN**

8. What remains OPEN

1. **A001418(8)** — exact 8×8 map folding count (lit known only through $n = 7$).
2. **Map-folding asymptotics / closed form / P-time algorithm** — OPEN.
3. GATE-04 Object 1 meter — collection **LOCKED** (orthogonal track; no invented J).

9. Non-claims

- Not Clay-closed; not finished-ToE; GATE-04 LOCKED.
- Not RH / P≠NP / YM / Frankl / NS-solved / AC-solved.
- Not "map folding solved"; not A001418(8); no new OEIS records claimed.
- Finite / lit ledger only.

10. Reproduction

```
cd /workspace/experiments/band1/map_folding_2xn
.venv/bin/python verify_map_folding_2xn.py
```

Wall-clock for the full suite on the 2026-09-14 box run: ≈ 4.4 **s**.

References (selected)

1. OEIS A000136, A000682, A001415, A001418 (accessed 2026-09-14).
2. Koehler, J. Combin. Theory 5 (1968).
3. Lunnon, Math. Comp. 22 (1968); Computer J. 14 (1971).
4. Wikipedia: Map folding; MathWorld: Map Folding / Stamp Folding.
5. Hunter Hogan, `mapFolding` (PyPI / GitHub) — Lunnon algorithm implementation.
6. Robert Dickau, Folding a 2xn Sheet of Labeled Stamps (Wolfram Demonstrations).
7. Parent Band-1 note: `2026-09-11_Band1_Computational_Experiment_map_folding`.